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Aluminium 6063

Scope

Aluminium 6063 is a medium strength alloy commonly referred to as an architectural alloy. It is normally used in intricate extrusions. It has a good surface finish, high corrosion resistance, is suited to welding and can be easily anodised. Most commonly available as T6 temper, in the T4 condition it has good formability.

Application

This material is used for architectural applications, shop fittings, irrigation tubing, balustrading, window frames, extrusions and doors

Supplied Forms

Extrusions

Tube

Alloy Designations

Aluminium alloy 6063/6063A also corresponds to: AA6063, Al Mg0.7Si, GS10, AlMgSi0.5, A-GS, 3.32206, ASTM B210, ASTM B221, ASTM B241 (pipe-seamless), ASTM B345 (pipe-seamless), ASTM B361, ASTM B429, ASTM B483, ASTM B491, MIL G-18014, MIL G-18015, MIL P-25995, MIL W-85, QQ A-200/9, SAE J454, UNS A96063 and HE19.

Temper Types

The most common tempers for 6063 aluminium are:

6063 O

Soft / No heat treatment.

6063 T4

Solution heat treated and naturally aged to a substantially stable condition.

6063 T6

Solution heat treated and artificially aged.

Corrosion resistance

Aluminium 6063 provides excellent corrosion resistance due to its ability to form an oxide layer on the surface that protects it from further oxidation. It is one of the preferred materials for outdoor applications where the metal may be exposed to harsh condition such as high humidity or salty oceanic atmospheres.

Heat resistance

Aluminium 6063 has excellent heat resistance. It can withstand temperatures of up to 1000⁰C. it can be used in high-temperature environments such as aerospace components or industrial machinery parts that require long-term use at elevated temperature without corroding or losing structural integrity over time.

Machining

Machining aluminium alloys tends to be challenging because the alloys deform easily when subjected to high cutting speeds or forces. Special care must be taken when machining aluminium 6063 because of its higher level of hardness compared with pure aluminium alloys.

Consider using slow speeds and low levels of feed pressure along with coolant lubricants when cutting to reduce deformation and hardness.

Fabrication

- Solderability: Good
- Weldability - Gas: Excellent
- Weldability - Arc: Excellent
- Weldability - Resistance: Excellent
- Brazability: Excellent
- Workability - Cold: Average
- Machinability: Average

Chemical Compositions

Element	% Present
Manganese (Mn)	0.00 - 0.10
Iron (Fe)	0.00 - 0.35
Magnesium (Mg)	0.45 - 0.90
Silicon (Si)	0.20 - 0.60
Zinc (Zn)	0.00 - 0.10

Element	% Present
Titanium (Ti)	0.00 - 0.10
Chromium (Cr)	0.00 - 0.10
Copper (Cu)	0.00 - 0.10
Aluminium (Al)	Balance

Mechanical properties at room temperature

Property	Value
Proof Stress	160 MPa
Tensile Strength	195 MPa
Elongation	14%
Shear Strength	150 MPa
Hardness Vickers	80 HV

Reference data for some physical properties (for guidance only)

Property	Value
Density	2.70 Kg/m ³

Property	Value
Melting Point	600 °C
Thermal Expansion	$23.5 \times 10^{-6} /K$
Modulus of Elasticity	69.5 GPa
Thermal Conductivity	200W/m.K
Electrical Resistivity	$0.035 \times 10^{-6} \Omega .m$

Editor

thyssenkrupp Materials (UK) LtdCox's LaneCradley HeathWest MidlandsB64 5QU

Important Note

Information given in this data sheet about the condition or usability of materials respectively products are no warranty for their properties, but act as a description. The information, we give on for advice, comply to the experiences of the manufacturer as well as our own. We cannot give warranty for the results of processing and application of the products.

Quicklinks

[Aluminium Home Page](#)

[Aluminium Datasheets](#)

Other datasheets

In addition to the above, we also have datasheets that detail the mechanical properties and chemical composition of the following aluminium grades:

[1050](#)

[3103](#)

[5005](#)

[5083](#)

[5251](#)

[5454](#)

5754

6060

6005

6082